

Creating a Large-Scale Monster with Style

David Allen
david.allen@dneg.com
DNEG Animation
Canada

Pietro Materossi
pom@dneg.com
DNEG Animation
United Kingdom

Aled Jones
arj@dneg.com
DNEG Animation
Canada

Bamboo Wang
zhwa@dneg.com
DNEG Animation
Canada

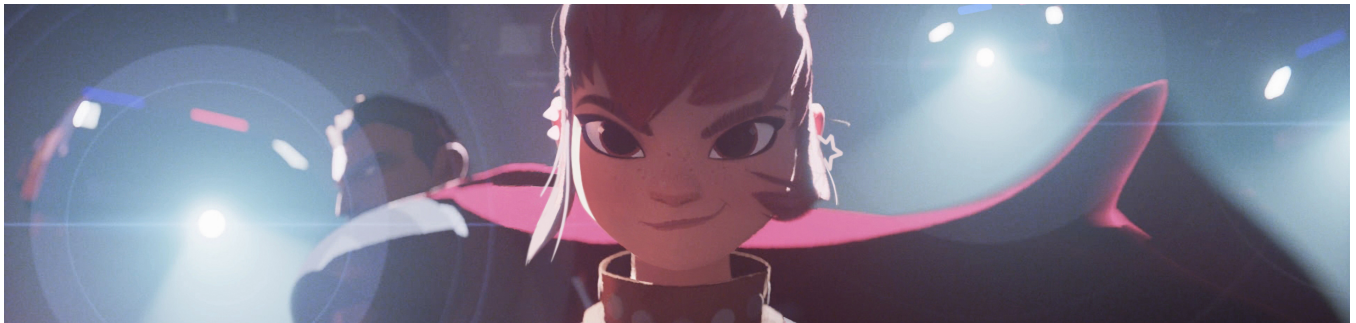


Figure 1: An unrelated image from the unreleased film "Nimona".

ABSTRACT

The creation of the stylized, large-scale monster in the film *Nimona* presented a number of unique visual and technical challenges. The creature was central to the story's emotional climax and needed to be highly emotive. The monster was portrayed on a grand scale, with highly stylized art direction and an abstract, amorphous design. The intention was to convey negative emotion using smoke-like streaming shapes. Here, we present an overview of the techniques used to bring this massive, dynamic, and highly stylized creature to life.

ACM Reference Format:

David Allen, Aled Jones, Pietro Materossi, and Bamboo Wang. 2023. Creating a Large-Scale Monster with Style. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '23 Talks)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587421.3595429>

1 INTRODUCTION

The monster needed to interact with its surroundings as it wreaked havoc on a futuristic cityscape. Since the monster's entire form was an always evolving depiction of stylized smoke, our approach

made heavy use of simulation in order to realize the highly dynamic nature of the monster's design.

The *Nimona* project was initially at BlueSky Studios, where a significant amount of visual experimentation of the monster had been done prior to DNEG's involvement. Using it as point of reference, we ultimately diverged from their look in order to achieve the expressiveness and performance.

The concept art of the monster was highly stylized, with a graphic style characterized by clean, flowing lines, and simple shapes. The design called for minimal shading and avoiding high-frequency detail. Despite these stylistic considerations, the monster was a behemoth, towering over the city at an imposing height of 200 meters. Our challenge was to stay true to the design aesthetic and still manage to convey a sense of immense scale.

2 MODELING, RIGGING, ANIMATION

Given that the effects would make up the bulk of the monster's form, the creature's model was kept simple, with a smooth, aerodynamic surface and minimal detail. There was no tail, as this would be generated by the effects department. The eyes were slits, represented by negative space. The rigging department was asked to minimize stretching and shearing of the base model and to provide proxy models of eye deformations for integration into the effects setup. Simple ribbon shapes were added to serve as pose guides for the shape of the streaming tendrils so animators could suggest effects performance.

Animators worked their magic, giving her skin surface and facial performance the personality, emotion, and believable scale appropriate weight as was needed to support this emotionally intense climax of the film. Special attention was paid to the eyes, as they

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH '23 Talks, August 06–10, 2023, Los Angeles, CA, USA
© 2023 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0143-6/23/08.
<https://doi.org/10.1145/3587421.3595429>

were an important device for conveying the monster's intense emotional state. Most of the creature's final appearance would come from the dynamics work done by effects artists.

3 SIMULATION

The effects team produced a variety of dynamic and procedural elements that were used to animate and realize the stylized treatment:

3.1 Volumetric Core

To define the monster's overall form, a volumetric core layer was derived from the skin surface performance. The density was carefully adjusted in different parts of the monster to define its overall opacity. The core was more dense and defined in the face area, with higher resolution to keep edges sharp and effectively convey facial animation. We then displaced the volume with noise according to hand-painted surface flow vectors. The volumetric dynamic smoke layer was derived from the animators' skin surface performance. The simulation used custom velocity fields to control the direction of flow, keeping it consistent with the flow of the other elements. Additional custom velocity fields were used to negate forces resulting from the monster's deformation. This helped tame the simulation as the monster moved through world space.

3.2 Procedural and Simulated Ribbons

Body ribbons were added as a procedural layer, using the animated skin as a guide. Artists painted curves in reference space to define the directional movement of the ribbons. These curves' tangents were adjusted to flow along the surface of the character's skin, then rasterized into a volume, serving as velocities for advection. This allowed for varying densities of ribbons to be created and advected along the surface based on the level of detail required for each shot. To represent stylized smoke tearing off the monster, a simulated dynamic ribbon layer was added. This layer depicted the smoke as stylized streaming geometric ribbons of variable density and size based on the distance from the camera and the desired level of detail. The ribbon layer was simulated as Vellum curves with dynamic constraints to control its behavior.

3.3 Facial Performance

To add life to the eye area, procedural ribbons with custom behavior were added around the monster's eyes. As the monster's eyes were critical to conveying the intense emotional state of the character, it was vital to retain the subtleties of the animators' performance. We achieved this by adding controls to the eye ribbons, making sure their motion never distracted attention.

To depict smoke tendrils tearing as the mouth opened, dynamic ribbons were added to represent the monster's teeth. The animation department had control over the placement and timing of the effect. The tearing effect was then achieved using procedurally generated curves with controls that allowed for breaking and tearing based on the proxy geometry.

3.4 Stylized Dust

A stylized dust effect was added using a simulated dynamic ribbon layer. This layer was designed to match the stylized ribbon treatment of the monster, but featured a much more explosive dynamic

with turbulent motion. This dust effect was mainly used to add impact to the monster's foot contact, when colliding with buildings, and when the monster was struck by projectiles.

4 RENDERING AND COMPOSITING

The setup generated several utility layers for review within the effects department, including a witness cam as a sanity check, as well as each layer rendered separately with holdouts. Automated Nuke scripts enabling effects artists to quickly render and evaluate the many simulated layers. The unique design of a smoke-like monster that absorbs light instead of emitting it, made the lighting process more streamlined. The primary source of lighting was the practical lights in the city environment, while special effects such as explosions, fire, and lasers were rendered as separate AOVs.

To create a sense of scale, the lighting artists emphasized the interaction between the monster and her surroundings through her inner-glowing eyes and heart. This served as the centerpiece, drawing the audience's attention and conveying the monster's emotions. The heart of the monster played a crucial role in the story, serving as the focal point that glowed from within the character. To achieve this, a dedicated light setup was created to track the monster's heart and adjust the lighting accordingly. This information was saved as metadata in the image files for use in compositing. The monster was then broken down into various elements and rendered with all necessary passes for the compositing team to assemble.

A defining characteristic of the monster's concept was the depiction of negative emotion through the absence of light, making it essentially a black hole of negative space. Consequently, the entire character's color palette was mostly black, occupying the bottom 5% of the color gamut in terms of value, which posed a challenge to the compositing team. To realize the final look, the team used atmosphere to help define its shape, rather than surface lighting. By utilizing position and surface data, comp artists graded pockets of atmosphere around the monster in favor of grading the monster itself. A vital style factor for the film was to avoid soft gradients in order to gain a more illustrative and graphic look. To simplify the tones, we utilized Cryptomatte IDs and passed the RGB values through a BlinkScript, initially developed by BlueSky Studios, to average out the values per ribbon.

5 CONCLUSIONS AND FUTURE WORK

We found the hardest part of this project was finding a look for the monster that not only satisfied the directors' vision, but also allowed us to create believable simulations. The initial design called for a nondescript form for the monster, but that look prevented the character's acting and expressiveness from showing through.

In the future, we'd like more finished animation for testing while effects are still in development. It would also be helpful to have lighting artists involved earlier, enabling us to show dev work in the proper context.